



مراجعة

العودة >

01

.d

00

.Your answer is incorrect

The correct answer is: 01

Incorrect

Mark 0.00 out of 5.00

سؤال 2

Say that $a(7 \text{ downto } 0)$ "11110011" and $b(3 \text{ downto } 0) = "1001"$. Determine the values produced by the assignments below

;($b(5 \text{ DOWNT0 } 2)$ XNOR (NOT b

الجواب:



0101

The correct answer is: 1010

Partially correct

Mark 13.13 out of 15.00

سؤال 3

KEY0 is resetn, KEY1 is the clock for --
req A. Operation: set SW to the value desired for A.





مراجعة

العودة >

Incorrect

Mark 0.00 out of 10.00

سؤال 5

To design a counter that counts the even numbers from ? 0 to 20, how many D-flip flops we will need
:Select one



.a
6



.b
5



.c
3



.d
4

.Your answer is incorrect
The correct answer is: 4

Correct

Mark 5.00 out of 5.00

سؤال 6

:The following VHDL code is for
entity XXX is





مراجعة

العودة >

```

; CLK<= 1
;wait Clock_Cycle/2
end loop

```

```

;END PROCESS
;END part4_arch

```

:Select one



.a

0000 1110



.b

0000 0000



.c

0010 1011



.d

0111 1011



.e

1001 0011

.Your answer is incorrect

The correct answer is: 1011 0111

Correct

Mark 10.00 out of 10.00

سؤال 9

Given the signals below





Correct

Mark 10.00 out of 10.00

سؤال 9

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0

;(signal Y : unsigned(7 downto 0

Which of the following is a legal VHDL assignment

:Select one



.a

;(Y <=to_unsigned(201, 8



.b

;(X <=unsigned(Y



.c

;(X <=to_integer(Y



.d

;(Y <=std_logic_vector(X

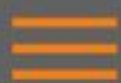
.Your answer is correct

;(The correct answer is: Y <=to_unsigned(201, 8

Partially correct

Mark 0.56 out of 15.00

سؤال 10





;(Y <=std_logic_vector(X

.Your answer is correct

;(The correct answer is: Y <=to_unsigned(201, 8

Partially correct

Mark 0.56 out of 15.00

سؤال 10

To test a 2x4 Decoder using Records, complete the following record declaration and its constant of :array

type test_vector is record

;(✓ downto 0 3)d : std_logic_vector

;(✗ downto 0 3) o : std_logic_vector

;end record

type test_vector_array is array (to 4 0 ✗) of

;test_vector

) =: constant test_vectors : test_vector_array

, ✗ , ✗ , ✗ , ✗)

, (✗ , ✗

, ✗ , ✗ , ✗ , ✗)





type test_vector is record

;(✓) d : std_logic_vector

;(✗) o : std_logic_vector

;end record

type test_vector_array is array () of

;test_vector

) =: constant test_vectors : test_vector_array

, ✗ , ✗ , ✗ , ✗)

, (✗ , ✗)

, ✗ , ✗ , ✗ , ✗)

, (✗ , ✗)

, ✗ , ✗ , ✗ , ✗)

, (✗ , ✗)

, ✗ , ✗ , ✗ , ✗)

;((✗ , ✗)



.Your answer is partially correct

.You have correctly selected 1

:The correct answer is

To test a 2x4 Decoder using Records, complete the following record declaration and its constant of :array

type test_vector is record

;([d : std_logic_vector([3 downto 0

;([o : std_logic_vector ([1 downto 0

;end record

type test_vector_array is array ([0 to 5]) of
;test_vector

) =: constant test_vectors : test_vector_array

,(['0'], ['0'], ['1'], ['0'], ['0'], ['0'])

,(['1'], ['0'], ['0'], ['1'], ['0'], ['0'])

,(['0'], ['1'], ['0'], ['0'], ['1'], ['0'])

;((['1'], ['1'], ['0'], ['0'], ['0'], ['1'])





D flip Flop with Asynchronous clear and enable



.d

D flip Flop with Synchronous reset



.e

D latch with enable



.f

D- Flip flop without clear, reset and enable

.Your answer is correct

The correct answer is: D flip Flop with Asynchronous clear

Correct

Mark 5.00 out of 5.00

سؤال 7

In the FPGA design flow, Timing based simulation occurs before synthesis

:Select one



True



False

.The correct answer is 'False

Incorrect





Mark 0.00 out of 10.00

Given the Testbench code bellow that aim to test a 8-bit shift register with serial input D, Clear input (SCLR operate on active high), and a clock (CLK). This shift register operate on +ev edge. Then what are the content of this register would be after 8 clock cycles :with shift left

```

                                ;LIBRARY ieee
                                ;USE ieee.std_logic_1164.all

                                ENTITY part4_vhd_tst IS
                                ;END part4_vhd_tst

                                ARCHITECTURE part4_arch OF part4_vhd_tst IS
                                ;Clock_Cycle: time := 40ns
                                ;SIGNAL Clk : STD_LOGIC

                                ;SIGNAL SCLR : STD_LOGIC
                                ;SIGNAL D : STD_LOGIC

                                SIGNAL O : STD_LOGIC

                                COMPONENT 4SRegister
                                ) PORT
                                ;Clk D, SCLR: IN STD_LOGIC
                                O : OUT STD_LOGIC
                                ;(

                                ;END COMPONENT

                                BEGIN

```





مراجعة

العودة



;

init : PROCESS
BEGIN

;SCLR<='0

;D<='0

;WAIT 10ns

;D<='1

;WAIT 15ns

;D<='0

;WAIT Clock_Cycle

;D<='1

;WAIT 2*Clock_Cycle

;D<='0

;WAIT Clock_Cycle

;D<='1

;WAIT 3*Clock_Cycle

;D<='0

;WAIT Clock_Cycle

;SCLR<='1

;WAIT Clock_Cycle

;WAIT

;END PROCESS init

Clock_1 : PROCESS

BEGIN

Clock_Loop: Loop

;Clk<= '0



```
BEGIN
    Clock_Loop: Loop
        ;'Clk<= '0
        ;wait Clock_Cycle/2
        ;'Clk<= '1
        ;wait Clock_Cycle/2
    end loop
;END PROCESS
;END part4_arch
```

Select one

- ☒ ☐
- a

0000 1110
- b

0000 0000
- c

0010 1011
- d

0111 1011
- e

1001 0011

.Your answer is incorrect



Correct

Mark 5.00 out of 5.00

سؤال 6

:The following VHDL code is for

entity XXX is

;port (B, D, R : in STD_LOGIC

;(Q: out STD_LOGIC

;end XXX

architecture arch of XXX is

begin

(X: process (B, R

begin

if (R='1') then

;'Q <= '0

else

if (rising_edge(B)) then

;Q <= D

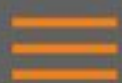
;end if

;end if

;end process X

;end arch

:Select one





;end process X

;end arch

:Select one



.a

D flip Flop with Asynchronous clear



.b

D flip Flop with Synchronous reset and enable



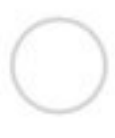
.c

D flip Flop with Asynchronous clear and enable



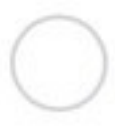
.d

D flip Flop with Synchronous reset



.e

D latch with enable



.f

D- Flip flop without clear, reset and enable

.Your answer is correct

The correct answer is: D flip Flop with Asynchronous clear

Correct

Mark 5.00 out of 5.00

سؤال 7





Incorrect

Mark 0.00 out of 10.00

سؤال 5

To design a counter that counts the even numbers from ? 0 to 20, how many D-flip flops we will need
:Select one

.a
6.b
5.c
3.d
4

.Your answer is incorrect

The correct answer is: 4

Correct

Mark 5.00 out of 5.00

سؤال 6





مراجعة

العودة

signal Carry : std_logic_vector (

;(✖ data_size-2 downto 0

begin

Full-Adder: for i in to data_size 0 ✔

generate

one_bit: FA

port map (A => (A(i) ✔ , B => (B(i) ✔ ,

Cin => (Carry(i) ✔ , Co =>

;(✔ (Sum(i) <= (Carry(i+1) ✔ , S

;end generate

;Cin => ✔ (Carry(0

; ✔ (Carry(data_size+1 => Cout

;end arch





Consider the full adder shown in the figure bellow. Select the right choices from the drop-list of the code top-level design in VHDL in order to have **8-bit full adder**

entity N-bit-FA is

;(generic (data_size : integer := 7

port (A, B : in STD_LOGIC_VECTOR(

;(✓ ▼ data_size downto 0

Sum : in STD_LOGIC_VECTOR(

;(✓ ▼ data_size downto 0

;(Cin, Cout: out STD_LOGIC

;end N-bit-FA

architecture arch of shreg is

COMPONENT FA IS

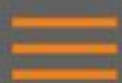
;PORT (A, B, Cin : IN STD_LOGIC

;(Co, S : OUT STD_LOGIC

;END COMPONENT

signal Carry : std_logic_vector (

;(✗ ▼ data_size-2 downto 0





Partially correct
Mark 13.13 out of 15.00

سؤال 3

KEY0 is resetn, KEY1 is the clock for --
reg_A. Operation: set SW to the value desired for A,
then store this value in A_reg by using KEY1. A is
shown on HEX3 and HEX2. Then change SW to the
value desired for B, which is displayed on HEX1 and
HEX0. The sum $S = A + B$ is displayed on HEX5 and
HEX4, and the carry out is shown on LEDR0. Notice
the the 7-segments leds turns on when signal 0 is
.given

```

;LIBRARY ieee
;USE ieee.std_logic_1164.all
;USE ieee.std_logic_unsigned.all

```

```

ENTITY part5 IS
;(PORT ( SW : IN STD_LOGIC_VECTOR(7 DOWNT0 0
;(KEY : IN STD_LOGIC_VECTOR(1 DOWNT0 0
;(LEDR : OUT STD_LOGIC_VECTOR(9 DOWNT0 0
HEX5, HEX4, HEX3, HEX2, HEX1, HEX0 : OUT
;((STD_LOGIC_VECTOR(0 TO 6
;END part5

```

```

ARCHITECTURE Behavior OF part5 IS
COMPONENT hex7seg
;(PORT ( hex : IN STD_LOGIC_VECTOR(3 DOWNT0 0
;((display : OUT STD_LOGIC_VECTOR(0 TO 6
;END COMPONENT
COMPONENT regn
;(PORT ( R : IN STD_LOGIC_VECTOR(7 DOWNT0 0
;Clock, Resetn : IN STD_LOGIC

```





```

;END COMPONENT
COMPONENT regn

```

```

;(PORT ( R : IN STD_LOGIC_VECTOR(7 DOWNT0 0

```

```

;Clock, Resetn : IN STD_LOGIC

```

```

;((Q : OUT STD_LOGIC_VECTOR(7 DOWNT0 0

```

```

;END COMPONENT

```

```

;(SIGNAL A, B : STD_LOGIC_VECTOR(7 DOWNT0 0

```

```

;(SIGNAL S : STD_LOGIC_VECTOR(8 DOWNT0 0

```

```

;SIGNAL carry : STD_LOGIC

```

```

BEGIN

```

```

(regn (R, Clock, Resetn, Q --

```

```

;(A_reg: regn PORT MAP (SW, KEY(1), KEY(0), A

```

```

;B <= SW

```

```

drive the displays through 7-seg decoders --

```

```

digit_3: hex7seg PORT MAP (A (DOWNT0 4 7) ✓

```

```

;(:, HEX3

```

```

digit_2: hex7seg PORT MAP (A (DOWNT0 0 3) ✓

```

```

;(:, HEX2

```

```

digit_1: hex7seg PORT MAP (B (DOWNT0 4 7) ✓,

```

```

;(HEX1

```

```

digit_0: hex7seg PORT MAP (B (DOWNT0 0 3) ✓

```

```

;(:, HEX0

```

```

; ✓ (A) + ('0' & B & '0') => S

```

```

;(carry <= S(8

```

```

digit_5: hex7seg PORT MAP (S (DOWNT0 4 7) ✓

```





```
; (, HEX5
```

```
digit_4: hex7seg PORT MAP (S (DOWNTO 0 3) ✓
```

```
; (, HEX4
```

```
; LEDR(0) <= carry
```

```
; "LEDR(9 DOWNT0 1) <= "000000000
```

```
; END Behavior
```

```
; LIBRARY ieee
```

```
; USE ieee.std_logic_1164.all
```

```
ENTITY regn IS
```

```
; (PORT ( R : IN STD_LOGIC_VECTOR(7 DOWNT0 0
```

```
; Clock, Resetn : IN STD_LOGIC
```

```
; ((Q : OUT STD_LOGIC_VECTOR(7 DOWNT0 0
```

```
; END regn
```

```
ARCHITECTURE Behavior OF regn IS
```

```
BEGIN
```

```
(PROCESS (Clock, Resetn
```

```
BEGIN
```

```
IF (Resetn = '0') THEN
```

```
; ('Q <= (OTHERS => '0
```

```
ELSIF (Clock'EVENT AND Clock = '1') THEN
```

```
; Q <= R
```

```
; END IF
```

```
; END PROCESS
```

```
; END Behavior
```

```
; LIBRARY ieee
```

```
; USE ieee.std_logic_1164.all
```

```
ENTITY hex7seg IS
```





```
                ;LIBRARY ieee
                ;USE ieee.std_logic_1164.all

                ENTITY hex7seg IS
;(PORT ( hex : IN STD_LOGIC_VECTOR(3 DOWNT0 0
                ;((display : OUT STD_LOGIC_VECTOR(0 TO 6
                ;END hex7seg

                ARCHITECTURE Behavior OF hex7seg IS
                BEGIN

                (PROCESS (hex
                BEGIN
                CASE hex IS
                ;"WHEN "0000" => display <= "0000001
                ;"WHEN "0001" => display <= "1001111
                ;"WHEN "0010" => display <= "0010010
                ;"WHEN "0011" => display <= "0000110
                ;"WHEN "0100" => display <= "1001100
                ;"WHEN "0101" => display <= "0100100
                ; x "1011111" => WHEN "0110" => display
                ;"WHEN "0111" => display <= "0001111
                ;"WHEN "1000" => display <= "0000000
                ;"WHEN "1001" => display <= "0000100
                ;"WHEN "1010" => display <= "0001000
                ;"WHEN "1011" => display <= "1100000
                ;"WHEN "1100" => display <= "0110001
                ;"WHEN "1101" => display <= "1000010
                ;"WHEN "1110" => display <= "0110000
                ;"WHEN OTHERS => display <= "0111000
                ;END CASE
                ;END PROCESS
                ;END Behavior
```





The truth table below represents a FSM with a 1-bit input (x) and a 1-bit output . Suppose the machine is initialized to its starting state '00' and then processes the input values: 1,0,0,1,1,1 (1 input bit after each Clk). What are the final state

Present State		Next State				Output	
		x = 0		x = 1		x = 0	x = 1
		A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

:Select one

☐

a.
11

☒

b.
10

☐

c.
01

☐

d.
00

.Your answer is incorrect





Present State		$x = 0$		$x = 1$		$x = 0$	$x = 1$
A	B	A	B	A	B	y	y
0	0	0	0	0	1	0	0
0	1	0	0	1	1	1	0
1	0	0	0	1	0	1	0
1	1	0	0	1	0	1	0

:Select one

☐

a.
11

☒

b.
10

☐

c.
01

☐

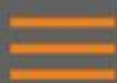
d.
00

.Your answer is incorrect
The correct answer is: 01

Incorrect
Mark 0.00 out of 5.00

سؤال 2

Say that a(7 downto 0) "11110011" and b(3 downto 0)





مراجعة

العودة >

.Your answer is incorrect

The correct answer is: 01

Incorrect

Mark 0.00 out of 5.00

سؤال 2

Say that $a(7 \text{ downto } 0) = "11110011"$ and $b(3 \text{ downto } 0) = "1001"$. Determine the values produced by the assignments below

$;(b(5 \text{ DOWNTTO } 2) \text{ XNOR } (\text{NOT } b$

الجواب:

0101



0101

The correct answer is: 1010

Partially correct

Mark 13.13 out of 15.00

سؤال 3

KEY0 is resetn, KEY1 is the clock for --
reg_A. Operation: set SW to the value desired for A,
then store this value in A_reg by using KEY1. A is
shown on HEX3 and HEX2. Then change SW to the
value desired for B, which is displayed on HEX1 and
HEX0. The sum $S = A + B$ is displayed on HEX5 and
HEX4, and the carry-out is shown on LED0. Notice





```
end loop  
  
;END PROCESS  
;END part4_arch  
:Select one
```



a. 0000 1110



b. 0000 0000



c. 0010 1011



d. 0111 1011



e. 1001 0011

.Your answer is incorrect
The correct answer is: 1011 0111

Correct
Mark 10.00 out of 10.00

سؤال 9

Given the signals bellow

```
;(signal X : std_logic_vector(7 downto 0
```





end loop
;END PROCESS
;END part4_arch
:Select one



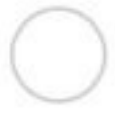
a.
0000 1110



b.
0000 0000



c.
0010 1011



d.
0111 1011



e.
1001 0011

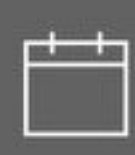
.Your answer is incorrect
The correct answer is: 1011 0111

Correct
Mark 10.00 out of 10.00

سؤال 9

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0





;END PROCESS
;END part4_arch
:Select one



.a
0000 1110



.b
0000 0000



.c
0010 1011



.d
0111 1011



.e
1001 0011

.Your answer is incorrect
The correct answer is: 1011 0111

Correct
Mark 10.00 out of 10.00

سؤال 9

Given the signals bellow

;(signal X : std_logic_vector(7 downto 0

;(signal Y : unsigned(7 downto 0





مراجعة

العودة >

```
;END PROCESS  
;END part4_arch  
:Select one
```



.a

0000 1110



.b

0000 0000



.c

0010 1011



.d

0111 1011



.e

1001 0011

.Your answer is incorrect

The correct answer is: 1011 0111

Correct

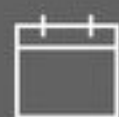
Mark 10.00 out of 10.00

سؤال 9

Given the signals bellow

```
;(signal X : std_logic_vector(7 downto 0
```

```
;(signal Y : unsigned(7 downto 0
```





Incorrect

Mark 0.00 out of 10.00

سؤال 8

Given the Testbench code bellow that aim to test a 8-bit shift register with serial input D, Clear input (SCLR operate on active high), and a clock (CLK). This shift register operate on +ev edge. Then what are the content of this register would be after 8 clock cycles :with shift left

```

                                ;LIBRARY ieee
                                ;USE ieee.std_logic_1164.all

                                ENTITY part4_vhd_tst IS
                                ;END part4_vhd_tst
                                ARCHITECTURE part4_arch OF part4_vhd_tst IS
                                ;Clock_Cycle: time := 40ns
                                ;SIGNAL Clk : STD_LOGIC

                                ;SIGNAL SCLR : STD_LOGIC
                                ;SIGNAL D : STD_LOGIC

                                SIGNAL O : STD_LOGIC

                                COMPONENT 4SRegister
                                ) PORT
                                ;Clk D, SCLR: IN STD_LOGIC
                                O : OUT STD_LOGIC
                                ;(

```





```

        COMPONENT 4SRegister
            ) PORT
;Clk D, SCLR: IN STD_LOGIC
        O : OUT STD_LOGIC
            ;(
;END COMPONENT

        BEGIN

            i1 : 4SRegister
            ) PORT MAP
list connections between master ports and --
            signals
            ,Clk => Clk
            ,D => D
            ,SCLR => SCLR
            O => O
            ;(

            init : PROCESS
                BEGIN

                    ;'SCLR<='0
                    ;'D<='0
                    ;WAIT 10ns
                    ;'D<='1
                    ;WAIT 15ns
                    ;'D<='0
                    ;WAIT Clock_Cycle
                    ;'D<='1

```





```

init : PROCESS
    BEGIN

        ;'SCLR<='0
        ;'D<='0
        ;WAIT 10ns
        ;'D<='1
        ;WAIT 15ns
        ;'D<='0
        ;WAIT Clock_Cycle
        ;'D<='1
        ;WAIT 2*Clock_Cycle
        ;'D<='0
        ;WAIT Clock_Cycle

        ;'D<='1
        ;WAIT 3*Clock_Cycle

        ;'D<='0
        ;WAIT Clock_Cycle

        ;'SCLR<='1
        ;WAIT Clock_Cycle
        ;WAIT
    ;END PROCESS init

Clock_1 : PROCESS

    BEGIN

        Clock_Loop: Loop
            ;'Clk<= '0
            ;wait Clock_Cycle/2
            ;'Clk<= '1
    
```



```
;WAIT Clock_Cycle  
;WAIT  
;END PROCESS init
```

```
Clock_1 : PROCESS  
BEGIN
```

```
Clock_Loop: Loop  
; 'Clk<= '0  
;wait Clock_Cycle/2  
; 'Clk<= '1  
;wait Clock_Cycle/2  
end loop
```

```
;END PROCESS  
;END part4_arch
```

:Select one



.a
0000 1110



.b
0000 0000



.c
0010 1011



.d
0111 1011



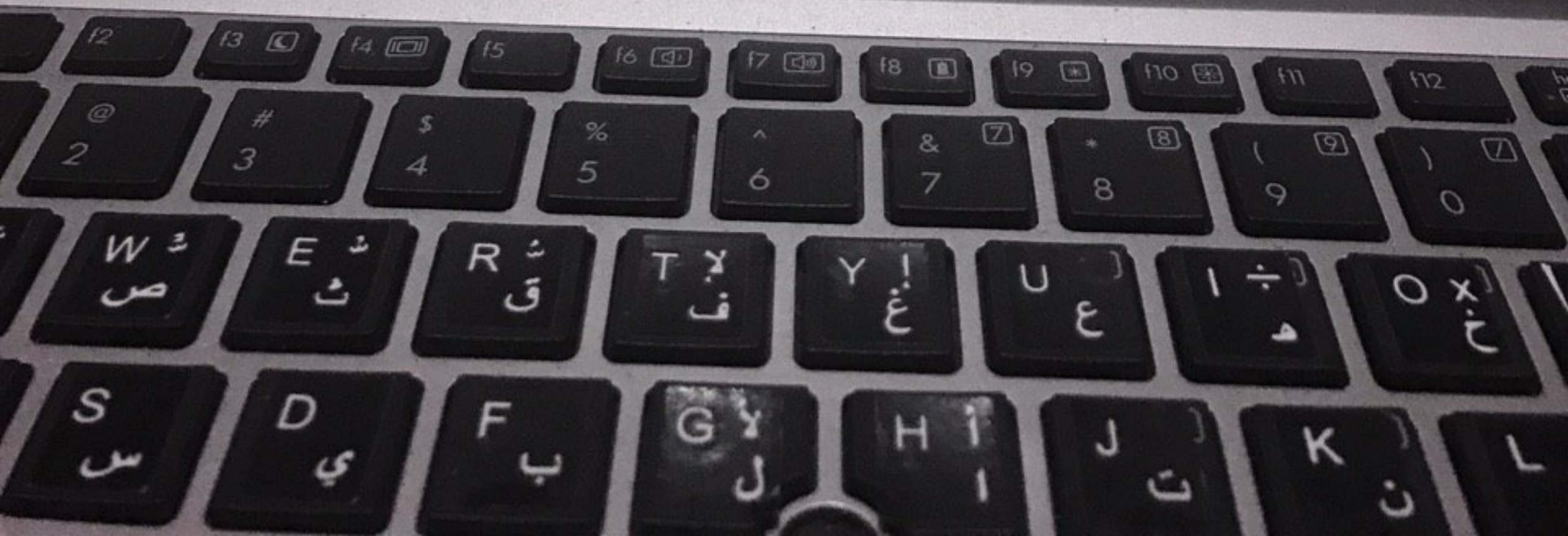
Time left 0:05:40

To design a counter that counts the even numbers from 0 to 20, how many D-flip flops we will need ?

Select one:

- ☐ a. 6
- ☐ b. 5
- ☐ c. 3
- ☐ d. 4

Next page



signal Carry : std_logic_vector (0 to data_size);

begin

Full-Adder: for i in 0 to data_size generate

one_bit: FA

port map (A => A(i), B => B(i), Cin => Carry(i)
=> Sum(i));

end generate;

Carry(0) <= Cin;

Cout <= Carry(data_size+1);

end arch;